

Faculty of Information Technology (FIT) – Ho Chi Minh City University of Science (HCMUS)

CS423 / CSC13003 – Software Testing (AI-augmented · 2026)

AI POLICY · TEMPLATES — 2026 v1.0

AI Audit Report — 5-section Template per Artifact

Mandatory appendix for every AI-assisted homework (HW#01–HW#06, and Seminar).

Adapted from Med Kharbach, PhD (2026) — AI Use Policy Templates for Higher Education. CC BY-NC-SA 4.0. This adaptation is prepared for FIT@HCMUS – CS423 / CSC15003 Software Testing course.

1. Student Information

Field	Value
Student name (printed):	Nguyễn Tấn Phát
Student ID:	23127449
Class / Cohort:	23KTPM2
Assignment ID:	HW#01
Assignment date:	2026-05-29
AI tool(s) used:	Gemini
AI tool(s) used:	☒ Yes ☐ No

2. Instructions (read before filling)

- Add one row per AI-generated artifact (test case, script, checklist, OpenAPI spec, JMeter plan, etc.).
- Paste the verbatim prompt — DO NOT paraphrase.
- Paste the verbatim AI output (or include a labelled screenshot in the report).
- Tag the verdict: VALID / INVALID / INCOMPLETE.
- Reasoning must cite a course slide, ISTQB section, or technical RFC.
- Show the corrected artifact with the change highlighted.
- Sample rows are in italic — replace them before submission.

3. Audit Table — one row per artifact

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
Artifact #1 Tool: Gemini Time: 2026-05-26 02:53:25 Prompt: Link to Prompt 1	Link to Output 1	VALID	The AI accurately extracted and summarized the assignment constraints, deliverables, and anti-cheat mechanisms from the provided documentation without hallucination.	Accepted as-is. Used as a baseline for the project workflow.

Artifact #2 Tool: Gemini Time: 2026-05-26 03:07:39 Prompt: Link to Prompt 2	Link to Output 2	VALID	Provided a technically sound search strategy and correctly advised on anti-cheat screenshot compliance and AI impact analysis structuring.	Accepted as-is. Applied the framework manually to gather the 10 job postings.
Artifact #3 Tool: Gemini Time: 2026-05-27 18:29:39 Prompt: Link to Prompt 3	Link to Output 3	INVALID	The AI committed semantic hallucination by generating fake 404 URLs. It also committed technical hallucination by fabricating a "blockchain" solution for the Chevrolet Tahoe exploit.	Manually researched and replaced all 20 URLs with genuine, working source links. Completely rewrote the Chevy Tahoe solution to reflect the factual incident response (disabling the bot).
Artifact #4 Tool: Gemini Time: 2026-05-28 02:57:17 Prompt: Link to Prompt 4	Link to Output 4	INCOMPLETE	While the standard functional paths (Happy Paths) were correct, the AI inherently lacks real-world physical context, leaving test coverage incomplete regarding hardware wear-and-tear and environmental interference.	Adopted the 12 base cases but manually engineered 3 physical edge cases (TC-13 Half-press, TC-14 Motor obstruction, TC-15 Soft surface vibration) to achieve comprehensive coverage.
Artifact #5 Tool: Gemini Time: 2026-05-28 12:43:55 Prompt: Link to Prompt 5	Link to Output 5	INVALID	Violates ISTQB Foundation Level syllabus. Assigned automation framework setup to Test Analyst, swapped Analysis/Design chronological order, and wrongly categorized Boundary Value Analysis as White-box.	Manually edited the Mermaid code. Reassigned roles correctly, fixed the Test Process order, and moved BVA to the Black-box branch.

4. Summary of AI Accuracy

Aggregate the verdicts from Section 3 and complete the table below.

Metric	Count	Percentage
Total AI-generated artifacts audited	5	100%

VALID (correct, accepted as-is)	2	40%
INVALID (wrong; rejected)	2	40%
INCOMPLETE (acceptable after edits)	1	20%

5. Conclusion — When should AI be used (or not)?

Write 80–150 words describing patterns you observed. Where did AI shine? Where did AI fail? What is your recommendation for using AI in this kind of work in the future?

Generative AI is highly effective for structuring large datasets, drafting boilerplate test cases, and summarizing software requirements where rapid data synthesis is needed. It significantly accelerates the initial phases of the Software Testing Life Cycle (STLC).

However, AI should **NOT** be used as a primary source for factual historical research, nor should it be trusted to design physical hardware testing scenarios without strict human oversight. AI lacks real-world physical context (e.g., it cannot inherently understand the mechanics of a tip-over safety switch on a soft mattress) and struggles with precise academic boundaries (such as strict ISTQB syllabus definitions). Therefore, AI must be utilized strictly as a drafting assistant. Factual verification, edge-case generation, and actual test execution must remain entirely human-driven to ensure product quality.

6. Mandatory Disclosure (paste verbatim)

"[Test cases / script / dataset / report] was initially generated by [AI tool name]; I reviewed and modified [section X], added [edge cases Y, Z]; [section W] was written entirely by me. The detailed AI Audit Report is attached as Appendix A. I confirm I did not use AI to generate any artifact listed in the prohibited category."

"[QA/QC job market analysis, software defect table, physical product test cases, and the ISTQB mindmap code] was initially generated by [Gemini]; I reviewed and modified [the hallucinated URLs, the Chevy Tahoe solution, and the ISTQB mindmap mistakes], added [edge cases TC-13, TC-14, TC-15]; [the actual physical testing execution videos, the AI Hallucination Analysis, and the AI Critique] was written entirely by me. The detailed AI Audit Report is attached as Appendix A. I confirm I did not use AI to generate any artifact listed in the prohibited category."

Signature

Student name (printed):	Nguyễn Tấn Phát
Student ID:	23127449
Class / Cohort:	23KTPM2
Course:	CS423 / CSC13003 – Software Testing
Instructor:	Dr. Lam Quang Vu / Dr. Tran Duy Hoang / MSc. Tran Thi Bich Hanh / MSc. Truong Phuoc Loc / MSc. Ho Tuan Thanh
Date:	2026-05-29
Signature:	

References

- Kharbach, M. (2026). AI Use Policy Templates for Higher Education. CC BY-NC-SA 4.0.
- ISTQB Foundation Level Syllabus (latest version).
- Hardman, P. (2025). A Post-AI Learning Taxonomy.
- Fuster Rabella, M. (2025). OECD Education Working Paper No. 338.
- Perkins, M., Roe, J., & Furze, L. (2025). AI Assessment Scale.
- Anthropic (2025). Building reliable AI test agents — engineering blog.
- DeepEval & Promptfoo documentation — testing frameworks for LLM systems.